

ANNOUNCING GLOBAL OPTIMISATION

19

► Sampling from π_β for large β is equivalent to global optimising

$$\propto \exp(-\beta U(x) - g(x)), \quad U(x) \equiv f(x) - \min_x f(x)$$

$$\pi_{\beta}(x) \propto \exp(-\beta f(x) - g(x))$$

► To see what happens as $\beta \rightarrow \infty$, note that we can write π_β as

As $\beta \rightarrow \infty$ we have the second term goes to zero and π_β concentrates on the null set of $\mathcal{U}$

$$= 1[U(x)=0] + 1[U(x)>0] \exp(-\beta U(x) - g(x))$$

$$\lim_{\beta \rightarrow \infty} \pi_\beta = \delta_{U=0}$$

► Note that $\mathcal{U}(x) \equiv 0$ is precisely when $f(x)$ is minimised

$$U(x) = 0 \iff x \in \arg \min_x f(x)$$

► Note that we did not make an assumption on the state-space

▶ Analyzing methods were first introduced to tackle hard combinatorial optimization problems

ANNEALING AS GLOBAL OPTIMISATION

19

- ▶ To see what happens as $\beta \rightarrow \infty$, note that we can write π_β as

$$\begin{aligned}\pi_\beta(x) &\propto \exp(-\beta f(x) - g(x)) \\ &\propto \exp(-\beta U(x) - g(x)), \quad U(x) = f(x) - \min_x f(x) \\ &= 1[U(x) = 0] + 1[U(x) > 0]\exp(-\beta U(x) - g(x))\end{aligned}$$

- ▶ As $\beta \rightarrow \infty$ we have the second term goes to zero and π_β concentrates on the null set of U

$$\lim_{\beta \rightarrow \infty} \pi_\beta = \delta_{U=0}$$

- ▶ Note that $U(x) = 0$ is precisely when $f(x)$ is minimised

$$U(x) = 0 \quad \Longleftrightarrow \quad x \in \arg \min_x f(x)$$

- ▶ Sampling from π_β for large β is equivalent to global optimising
- ▶ Note that we did not make an assumption on the state-space
- ▶ Annealing methods were first introduced to tackle hard combinatorial optimization problems

